A Project Report on

Mini Food Delivery Web Application Using Core Java

submitted to

OBJECT ORIENTED PROGRAMMING

(ELC434 P)

by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Academic Year: 2025-26

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that Mr./Ms. **Abhi Raksha, Niwin Pranav M V, Srishti S Rao**, bearing Register No. **2462601, 2462620, 2462623**, has satisfactorily completed the project work titled "Mini Food Delivery Web Application Using Core Java" in the laboratory course, ELC434 P- OBJECT ORIENTED PROGRAMMING prescribed by the Department of Electronics and Communication Engineering, for the IV semester B.Tech. program during the academic year 2025-26.

Praghash K	**Dr Inbanila**
Faculty In-charge	**Head of the Department**

Date: 27-02-2026

Place: Bangalore

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Vision

"Excellence and Service"

Mission

"CHRIST (Deemed to be University) is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment."

Core Values

Faith in God
Moral Uprightness
Love of Fellow Beings
Social Responsibility
Pursuit of Excellence

Department Vision

" To be a center of excellence in education, research, and innovation in the field of engineering, creating technically competent and socially responsible professionals. "

Department Mission

M1: *To impart high-quality technical education through an innovative and engaging teaching-learning process.*

M2: *To foster a culture of research, innovation, and product development by collaborating with industry.*

M3: *To equip students with practical skills in Signal Processing, Communication Systems, and Embedded Technologies.*

M4: *To inculcate professional ethics, leadership qualities, and a mindset for lifelong learning.*

M5: *To address real-world societal challenges through sustainable engineering solutions.*

Program Educational Objectives (PEOs)

PEO1: *Graduates will possess strong foundational knowledge to analyze and solve complex engineering problems.*

PEO2: *Graduates will successfully design and implement engineering systems and solutions.*

PEO3: *Graduates will exhibit effective communication, teamwork, and project management skills.*

PEO4: *Graduates will engage in lifelong learning to adapt to rapidly changing technological landscapes.*

PEO5: *Graduates will demonstrate ethical practices and a strong commitment to societal and environmental well-being.*

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1: Apply engineering principles to design and develop real-time applications.
PSO2: Utilize programming tools and software technologies for application development.
PSO3: Analyze system performance and optimize solutions efficiently.
PSO4: Develop scalable and maintainable software systems.
PSO5: Apply ethical and professional practices in engineering solutions.

Evaluation Rubrics

- Laboratory Marks (50M) = Lab Experiments (25M) + Lab Project (25M)
- The lab project project is assessed for 100 marks and scaled down to 25 Marks.
- The following evaluation rubrics followed for assessment:
  - Problem formulation: 10 marks
  - Literature survey: 10 marks
  - Project design or methodology: 10 marks
  - Implementation of project: 20 marks
  - Result analysis: 10 marks
  - Project impact: 5 marks
  - Cost estimation: 5 marks
  - Presentation skill: 5 marks
  - Team work: 5 marks
  - Novelty and Originality: 10 marks
  - Project report : 10 marks

No.	Component	Date	Marks	Signature
1	Problem formulation(10)			
	Literature survey(10)			
2	Project design or methodology (10)			
	Implementation of project (20)			
3	Result analysis (10)			
	Project impact (5)			
	Cost estimation (5)			
4	Presentation skill (5)			
	Team work (5)			
	Novelty and Originality (10)			
	Project report (10)			
Total (100)				
Total (25)				

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CHAPTER 1

Introduction

The Mini Food Delivery Web Application is developed using Core Java. It demonstrates Object-Oriented Programming concepts and REST API handling using Java's built-in HttpServer.

PROBLEM STATEMENT

To develop a lightweight web-based food ordering system that supports user authentication and order processing without using external frameworks.

OBJECTIVES

1. Implement OOP concepts.
2. Create REST APIs using HttpServer.
3. Perform query parsing using Map.
4. Send JSON responses to client.
5. Calculate order totals dynamically.

SCOPE OF THE PROJECT

- Educational demonstration of backend development.
- Understanding client-server communication.
- Foundation for advanced frameworks like Spring Boot.

CHAPTER 2

System Design and Methodology

SYSTEM ARCHITECTURE

The project follows Client-Server Architecture.

- Client sends HTTP request.
- Server parses query parameters.
- Business logic is executed.
- JSON response is sent back.

AUTHENTICATION LOGIC

Login validation is performed using:

$$Login = (Username == StoredUsername) \wedge (Password == StoredPassword) \tag{2.1}$$

ORDER CALCULATION

Total amount is calculated as:

$$Total = \sum_{i=1}^{n} Price_i \tag{2.2}$$

OOP CONCEPTS USED

Concept	Implementation
Inheritance	Customer extends User
Encapsulation	Private variables in Driver

Composition Order has Items

Collections ArrayList

CHAPTER 3

Implementation and Results

MODULES DEVELOPED

- User.java
- Customer.java
- Item.java
- Driver.java
- Order.java
- WebServer.java

SAMPLE CODE SNIPPETS

3.2.1 Login Validation

```
public boolean validateLogin(String u, String p) {
    return username.equals(u) && password.equals(p);
}
```

3.2.2 Query Parsing

```
String[] pairs = query.split("&");
String[] keyValue = pair.split("=");
```

3.2.3 JSON Response

```
exchange.getResponseHeaders().set("Content-Type","application/json");
```

RESULTS

Test Case	Result
Valid Login	Success
Invalid Login	Fail
Order Placement	Total Generated

The system successfully:

- Authenticates users
- Parses query parameters
- Calculates total bill
- Sends JSON responses

CHAPTER 4

References

1. Oracle Java Documentation.
2. Java Collections Framework Guide.
3. REST API Design Principles.
4. HttpServer Documentation (com.sun.net.httpserver).